

MyDesign Portfolio

Element A Initial Template

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

First Poster:

Scenario

We have been asked to create a This or That design game for the middle school library. In line, kids can become bored waiting, so we are creating an amusing and entertaining game that can spark conversation and pass time. Some of our bigger constraints were durability and being able to fit on the library table.

Brainstorming designs

Our team came up with six designs. Here are the two that were most popular aside from the chosen design:
<https://docs.google.com/document/d/16a3SMX1AWipVvpjZ4xbXIXbPH3uXPo1QFRiVh9Maby8/edit?usp=sharing>

Decision Matrix

Based off of our criteria and constraints, we created a weighted decision matrix to score our designs.

	Weight	Sofia's First Design	Sofia's Ballot Design	John's First Design	John's Second Design	James' Design	Matthew's Design
Engagement/Interactivity	0.2	4	4	3.8125	3.5625	3.625	3.375
Cost/Construction	0.09	3	3.575	3.5	2.825	3	3.5
Durability	0.18	4	3.875	3.375	3.5	3.0625	3.75
Ease of Resetting/Use	0.21	3.8125	3.8125	3.875	3.625	3.375	3.625
Functional Performance	0.17	4	4	3.5	3.875	3.875	3.5
Safety	0.15	4	4	3.875	3.75	3.875	3.9375
Score		22.8125	23.2625	21.9375	21.1375	20.8125	21.6875

Design

We chose Sofia's ballot design. The ballot idea came from making kids feel that their choice helped get one of the options to win.

The majority of the design will be made will finished wood. The plexiglass in the front is for the kids to see their choice making a difference right away.

The librarian wanted to design to be easily reset, so the whiteboard was something that was easy to write and remove.

Testing

We created a testing plan to assess our design based on our criteria. Here is a link to the full testing plan:
<https://docs.google.com/document/d/1UGJvEHNdwZik9SpgDtzOpURTTGldAbcsh3vRT-lbqU/edit?usp=sharing>

Requirement	Testing Setup and Procedure
Ease of use/resetting	Test out the design to see if it is simple to complete and not complicated. Ask someone, preferably younger to test and see if they can easily figure it out. Determine the results once they have done so.
Engagement/Interactivity	Ask the people from before if they think the design is engaging and could be a conversation starter for middle schoolers.
Durability	Drop from three times from different heights and see if the design breaks or falls apart.
Safety	Should be safe. Check the edges of the design for corners that could injure the kids. Examples of this could be splinters, cuts, bruises, etc.

Testing Results

Here are the results of our testing:

	Result			
	Sofia	James	Matthew	John
Ease of use/resetting	Goal	Goal	Goal	Goal
Engagement/Interactivity	Pass	Goal	Goal	Pass
Durability	Goal	Pass	Pass	Goal
Safety	Goal	Goal	Goal	Goal

Material List

Wooden Boards
Wood glue
White Board
Marker
Plexiglass
Tokens

Presentation

Google Docs
This or That Presentation Video.mp4

Prototypes

Matthew's
Sofia's
James's
John's

Video of prototype

Onshape Drawing

Sketch

THE CAD-ASTROPHES

Where Ctrl+Z is our most used tool.

This or That Presentation Video First Iteration.mp4

Prototype design v1 video.MOV

Second Poster:

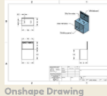
Name: John M.

Date: 5/17/26

Class: HS Engineering

Original Design

The majority of the design will be made with finished wood. The plexiglass in the front is for the kids to see their choice making a difference right away. The librarian wanted the design to easily reset, so the whiteboard was something that was easy to write and remove.

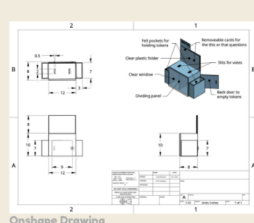
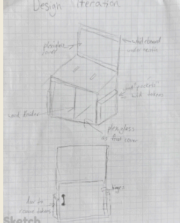


Onshape Drawing Sketch Prototype

Design Iteration

Here is our modified design after feedback:

Similar to the original design, but with pockets on the side to hold the tokens before voting. The back will have a door cut out on the back for easy access to remove the tokens. It will allow the kids to pull out the wooden tokens and keep using them for future voting. The pockets on the sides will be made of felt for ease of use and not to add too much weight to the design. The top of the back board will have a clear folder that will hold two papers cards that can be slide out and changed easily. The box will be painted white, but as the kids vote and choose, they can draw and write on it to customize.



Sketch Onshape Drawing

Feedback Chart

Here is our feedback from the initial board. Here are four suggestions we took into account.

Feedback	Thoughts	Modifications
How will they remove the votes?	We agree that there needs to be a way to remove the votes.	We will add a wooden door and metal hinge at the back of the box to empty the tokens when the vote is over.
Where are the tokens kept?	This is an important aspect of our design that we were missing.	A felt pocket-like holder will be glued to each side of the box to store the tokens.
Eye catching graphics are needed to get students' attentions.	Indeed, the box needs color and decor.	The box will be painted white, but as the kids vote and choose, they can draw and write on it to customize.
Whiteboard is easily erasable and students may be tempted to erase it. Also, the location of the whiteboard may make for an awkward hand position.	We agree that a whiteboard can be messy and hard to write on, and that students may be tempted to erase it.	Instead of using a whiteboard, the questions will be displayed on paper cards. They will slot into a clear plastic folder glued to the front.

Testing

Requirement	Design, Build and Test Iteration	Results
Base of the	Tested the design to see if it is capable to complete and not complicated. Ask at least two other people, preferably younger, to test and give their opinion on how easy it was to complete.	Pass: The kids do not know what to do at all in how to proceed with the game. Fail: The kids know what to do to complete the game but struggle slightly to fully complete it. Goal: The kids know exactly how to put in their votes and how to complete the game.
Engagement/interactivity	Ask the people from before if they think the design is engaging and would be a conversation starter for middle students.	Pass: The kids are bored by the design on first understanding. Fail: The kids complete the game and find that they want talking about. Goal: The kids find the design really interesting and have a good conversation starting from the game.
Usability	Drop some items in the box, once from 3 feet, and once from 4 feet.	Pass: The design falls apart and cannot be salvaged. Fail: The design falls apart and cannot be salvaged. Goal: All parts of the prototype remain unchanged when the prototype is returned to the original position.
Safety	Should be safe. Check the edges of the design for corners that could injure the kids. Number of the number of corners, cuts, splinters, etc.	Pass: The design falls apart and cannot be salvaged. Fail: The design falls apart and cannot be salvaged. Goal: The design falls apart and cannot be salvaged.
Token Holder/Container	Should be able to hold 50 tokens on each side.	Pass: The design falls apart and cannot be salvaged. Fail: The design falls apart and cannot be salvaged. Goal: The design falls apart and cannot be salvaged.
Back Door	Drop the tokens through the slot. Open the back door, empty the tokens, and close the back door.	Pass: The design falls apart and cannot be salvaged. Fail: The design falls apart and cannot be salvaged. Goal: The design falls apart and cannot be salvaged.
Cards	Slide the cards in and out of the folder.	Pass: The design falls apart and cannot be salvaged. Fail: The design falls apart and cannot be salvaged. Goal: The design falls apart and cannot be salvaged.
Size	Measure the base of the device. The dimensions should be no larger than 11"x18".	Pass: The design falls apart and cannot be salvaged. Fail: The design falls apart and cannot be salvaged. Goal: The design falls apart and cannot be salvaged.

Presentation

Google Docs

This or That Presentation Second Iteration.mp4

Prototype Video

Google Docs

John This or That Second Iteration Protot...

Prototype Photos

These are the photos of our new prototypes.



Sofia James



John Matthew

Material List

- Wooden boards
- Wood glue
- Plexiglass
- Tokens
- Felt
- Metal hinges
- Clear folder
- Cards



📺 John This or That Second Iteration Prototype Video.mov

📺 This or That Presentation Second Iteration.mp4

We have been asked to create a This or That device for the Sky Ranch Middle School library. The device will be placed on the front desk of a library, and act as a way for kids to pass time in line, and perhaps spark conversation. The device works by having two options, and the kids would vote on which they would rather choose. The librarian had a working prototype, which consisted of two glass jars with cards taped to them that displayed the options. Kids would take pom-poms from a plastic bin in between the jars, and drop it in the jar they wanted to vote.

The librarian did not like that the parts were in separate pieces, and could see some parts falling on the floor. She also didn't like that there was glass, and could see it falling off the desk and shattering. So, she asked us to design something for her that would solve these

problems. She joined the class of the first section, and she described the problem a little more, as well as answered some questions the other section asked.

Some key constraints are that the base of the device must be no larger than 11"x18". The device should be in one piece, and should not contain any glass parts. The device should be sturdy and shouldn't fall off the desk. The this or that options should be easily replaceable, as the questions would cycle periodically.

Stakeholder Concerns and Connections (A3, A4)

Primary stakeholder groups are the librarian, and the kids attending the library. The kids will be most affected, as they will use the device. The library is what will be facilitating the this or that votes, and is the person who requested the device. She wanted an annual thing at the library, last time it was a LEGO tower, but this time she wants a this or that device. This project doesn't have many other stakeholders, as the recipient of the device is the librarian, and the kids will use the device. This project probably won't have any social or environmental impacts on secondary stakeholders, as this is a small, local problem and a small, insubstantial solution that will only affect the library and patrons of the library.

Problem Importance (A6)

The library wants a fun-interactive way for kids to pass time in line, as well as spark conversations.